

AP-E14 Wess–Zumino Flux Decay and Morrey Regularity: The Exact Hopf Base/Fibre Split and a Div–Curl No-Go

Route-F derivation ledger

20 July 2026

Abstract

This note executes the two continuations left by AP-E13. First, for every *fixed* complete-minor graph map $u : \Omega \rightarrow \mathbb{S}^3$, including the selected relaxed $B = 1$ representative, the local Wess–Zumino flux obeys

$$\sup_x \frac{\text{dist}(\int_{B_r(x)} u^* \omega_3, 2\pi^2 \mathbb{Z})^2}{r^3} \leq \frac{4\pi}{3} \sup_x \int_{B_r(x)} |M_3(Du)|^2 dx = o(1).$$

This is an unconditional consequence of L^2 absolute continuity and does not use minimality. It does not imply the linear target oscillation needed by the AP-E13 replacement: a localized circle-valued rank-one defect has $M_2 = M_3 = 0$, zero Wess–Zumino flux, finite Dirichlet energy, and oscillation diameter two at every scale. Hence the Morrey statement for the actual relaxed minimizer remains a genuinely variational regularity problem.

Second, writing $n = h(u)$ for the Hopf projection and $a = 2u^*\theta$ for the fibre one-form gives the exact identities

$$|Du|^2 = \frac{1}{4}(|a|^2 + |Dn|^2), \quad |M_2(Du)|^2 = \frac{1}{16}(|a \wedge Dn|^2 + |da|^2), \quad |M_3(Du)|^2 = \frac{1}{64}|a \wedge da|^2.$$

Only the signed Chern–Simons integral has exact differential cancellation. A periodic lift with $da = a \wedge da = 0$ but $\int |M_2(Du)|^2 > 0$ isolates the surviving vertical–horizontal shear and disproves a universal reduction of the full cutoff remainder to an exact div–curl pairing. A 12/12 verification card checks the algebra, counterexamples, exact local Wess–Zumino quadrature, and three independently relaxed compatible-action $B = 1$ grids. The finite-grid Morrey proxy improves, but it is not a continuum theorem. Hessian, determinant, and portal gates remain closed.

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1 Scope and conventions

Let $\Omega \subset \mathbb{R}^3$ and let $u : \Omega \rightarrow \mathbb{S}^3 \subset \mathbb{R}^4$ belong to the complete-minor graph class

$$u \in W^{1,2}, \quad M_2(Du), M_3(Du) \in L^2. \quad (1)$$

The derivative part of the Route-E continuum action is

$$W(Du) = \frac{1}{2}(|Du|^2 + R|M_2(Du)|^2 + K|M_3(Du)|^2), \quad R, K > 0. \quad (2)$$

We orient $\mathbb{S}^3$ as the boundary of the standard unit ball in $\mathbb{R}^4$ and write ω_3 for its volume form, so $\int_{\mathbb{S}^3} \omega_3 = 2\pi^2$. For an $\mathbb{S}^3$ -valued graph map,

$$u^* \omega_3 = J_u \, dx, \quad J_u = \det[u, \partial_1 u, \partial_2 u, \partial_3 u], \quad |J_u| \leq |M_3(Du)|. \quad (3)$$

All statements below distinguish a fixed graph representative from an arbitrary scale-dependent family. This distinction is precisely why the AP-E13 cap sequence is not a counterexample to the next theorem.

2 Unconditional Wess–Zumino flux decay for a fixed map

For $\Omega' \Subset \Omega$ and $0 < r < \text{dist}(\Omega', \partial\Omega)$ define the local relative class and its least representative by

$$\mathcal{V}_u(x, r) = \left[\int_{B_r(x)} J_u \, dy \right] \in \mathbb{R}/(2\pi^2\mathbb{Z}), \quad \delta_u(x, r) = \text{dist} \left(\int_{B_r(x)} J_u \, dy, 2\pi^2\mathbb{Z} \right). \quad (4)$$

Theorem 2.1 (Uniform fixed-map WZ-flux decay). *Every u satisfying (1) obeys*

$$\sup_{x \in \Omega'} \frac{\delta_u(x, r)^2}{r^3} \leq \frac{4\pi}{3} \sup_{x \in \Omega'} \int_{B_r(x)} |M_3(Du)|^2 \, dy \xrightarrow{r \downarrow 0} 0. \quad (5)$$

In particular this applies to any fixed relaxed $B = 1$ minimizer represented in the complete-minor graph class; neither minimality nor an Euler–Lagrange equation is used.

Proof. The zero lattice element is an admissible comparison in (4). Using (3) and Cauchy–Schwarz,

$$\delta_u(x, r)^2 \leq \left| \int_{B_r(x)} J_u \, dy \right|^2 \leq |B_r| \int_{B_r(x)} |M_3(Du)|^2 \, dy. \quad (6)$$

Since $|B_r|/r^3 = 4\pi/3$, this gives the inequality. Put $f = |M_3(Du)|^2 \in L^1(\Omega)$ and

$$\alpha_f(t) = \sup \left\{ \int_E f \, dy : E \subset \Omega \text{ measurable}, |E| \leq t \right\}.$$

Absolute continuity of the Lebesgue integral says $\alpha_f(t) \rightarrow 0$ as $t \downarrow 0$. Every $B_r(x)$ has measure $4\pi r^3/3$, uniformly in x , so the right-hand side of (5) is at most $(4\pi/3)\alpha_f(4\pi r^3/3) = o(1)$. $\square$

Remark 2.2 (Sharp logical scope). *The conclusion is $\delta_u(x, r) = o(r^{3/2})$, uniformly on compact subsets. AP-E13 instead chose a different filling candidate u_r at each shrinking scale and kept a fixed amount of $|M_3|^2$ inside the associated ball. Absolute continuity for one fixed L^1 density gives no uniform estimate over such a non-equintegrable family.*

The decay can be arbitrarily slow. For example the radial majorant

$$f(y) = \frac{\mathbf{1}_{|y| < 1}}{|y|^3 \log^2(e/|y|)} \quad (7)$$

is locally integrable and $\int_{B_r} f \, dy = 4\pi/\log(e/r)$. Thus no power improvement follows from L^2 control alone.

3 Why WZ decay does not give linear Morrey oscillation

The AP-E13 recovery lemma needs a normal-ball trace estimate of the form

$$\text{osc}_{B_r(x)} u \leq \Lambda r \quad (8)$$

or a comparable L^∞ bound. Graph energy, fixed degree, and (5) cannot imply it.

Proposition 3.1 (Rank-one zero-flux obstruction). *There exists an $\mathbb{S}^3$ -valued finite graph-energy map, localizable in a vacuum cylinder of a smooth compactly supported $B = 1$ map, for which $M_2 = M_3 = J_u = 0$ almost everywhere but the essential target oscillation has diameter two in every neighbourhood of the cylinder axis.*

Proof. In cylindrical coordinates (r, φ, z) take

$$u(r, z) = (\cos \ell(r), \sin \ell(r), 0, 0), \quad \ell(r) = \log \log(e/r), \quad 0 < r < r_0 < 1. \quad (9)$$

Because $u = \gamma \circ \ell$ for a circle-valued curve γ , Du has rank at most one; hence every second and third minor and the WZ density vanish. Per unit length of the axis,

$$\int_{0 < r < R} |Du|^2 dx = 2\pi \int_0^R \frac{dr}{r \log^2(e/r)} = \frac{2\pi}{\log(e/R)} \xrightarrow{R \downarrow 0} 0. \quad (10)$$

Nevertheless $\ell(r) \rightarrow \infty$ and traverses every phase infinitely often, so every axis neighbourhood has essential image diameter two.

To localize, replace ℓ by a scalar phase $\psi(r, z)$ which equals it in a smaller cylinder and vanishes near the cylinder boundary. The composite $\gamma \circ \psi$ remains rank one. Insert it where a smooth compactly supported degree-one map is constant. Smooth radial truncations of ψ have zero local degree and preserve the exterior degree, so the resulting graph limit has $B = 1$. $\square$

This proposition does *not* prove that the selected relaxed minimizer is irregular. It proves the sharper methodological statement that its minimality must be used. At present the relaxed functional is not known to admit the fixed-trace local recovery required to identify it with the naive integral (2); consequently inserting the formal classical Euler–Lagrange equation here would be circular.

4 Exact Hopf base/fibre decomposition

Write $u = (z, w) = (x_1 + ix_2, x_3 + ix_4) \in \mathbb{S}^3 \subset \mathbb{C}^2$ and define

$$h(u) = n = (2\Re(z\bar{w}), 2\Im(z\bar{w}), |z|^2 - |w|^2), \quad (11)$$

$$\theta = -x_2 dx_1 + x_1 dx_2 - x_4 dx_3 + x_3 dx_4, \quad a = 2u^* \theta. \quad (12)$$

For the standard orientation $\omega_2 = \frac{1}{2} \epsilon_{abc} n_a dn_b \wedge dn_c$ on $\mathbb{S}^2$, direct differentiation gives

$$da = -n^* \omega_2. \quad (13)$$

Under the fibre-coordinate change $u \mapsto e^{i\chi} u$,

$$n \mapsto n, \quad a \mapsto a + 2 d\chi. \quad (14)$$

This is a useful coordinate split, but not a gauge redundancy of the full Route-E model: the lift u is physical, and the mass term as well as the $|a|^2$ energy can depend on its fibre phase.

Theorem 4.1 (Complete-minor Hopf identities). *For every tangent derivative $Du \in \mathbb{R}^{4 \times 3}$,*

$$|Du|^2 = \frac{1}{4}(|a|^2 + |Dn|^2), \quad (15)$$

$$|M_2(Du)|^2 = \frac{1}{16}(|a \wedge Dn|^2 + |da|^2), \quad (16)$$

$$|M_3(Du)|^2 = \frac{1}{64}|a \wedge da|^2, \quad (17)$$

$$\det[u, Du] = \frac{1}{8} * (a \wedge da), \quad B = \frac{1}{16\pi^2} \int_{\Omega} a \wedge da. \quad (18)$$

The sign in (18) follows from the stated θ and the boundary orientation of $\mathbb{S}^3$; reversing either orientation reverses both displayed signs and changes no energy identity.

Proof. The identities are invariant under the transitive $U(2)$ action, so evaluate at $u = (1, 0, 0, 0)$. A tangent column has coordinates $p = (0, v, p_3, p_4)$. At this point

$$a(p) = 2v, \quad Dn(p) = (2p_3, -2p_4, 0), \quad da(p, q) = 4(p_3q_4 - p_4q_3). \quad (19)$$

The vertical line and horizontal two-plane are orthogonal, proving (15). For each pair of domain columns, the squared two-area is the sum of the vertical–horizontal parallelogram and horizontal determinant squares. Equation (19) multiplies each by sixteen, proving (16). The three-volume is the determinant of the one vertical and two horizontal components; $a \wedge da$ multiplies it by eight. This proves (17) and (18). $\square$

Consequently (2) becomes

$$W = \frac{1}{8}(|a|^2 + |Dn|^2) + \frac{R}{32}(|a \wedge Dn|^2 + |da|^2) + \frac{K}{128}|a \wedge da|^2. \quad (20)$$

The split is compatible with Sobolev Hopf-lifting theory [1, 4] and with recent special Faddeev–Skyrme minimality results [3], but those theorems do not supply L^2 control of every complete minor in the present $\mathbb{S}^3$ problem.

5 The exact div–curl test

The signed Chern–Simons density has the expected exact cancellation:

$$(a + 2d\chi) \wedge d(a + 2d\chi) = a \wedge da + 2d(\chi da). \quad (21)$$

Thus $\int a \wedge da$ is invariant on a torus or under fixed boundary data. The positive terms in (20) do not share this cancellation.

Proposition 5.1 (Chern–Simons exactness does not remove the cutoff stress). *The AP-E13 positive cutoff remainder cannot, in general, be rewritten only as an exact divergence or a curl-controlled pairing in the Hopf variables.*

Proof. Two independent periodic tests isolate the failure.

First let $T^3 = (\mathbb{R}/2\pi\mathbb{Z})^3$ and

$$u = \left(e^{i\chi} \cos \frac{t}{2}, e^{i\chi} \sin \frac{t}{2} \right), \quad t = A \sin x_1, \quad \chi = B \sin x_2. \quad (22)$$

Then $n = (\sin t, 0, \cos t)$ has rank one, $a = 2B \cos x_2 dx_2$, and

$$da = a \wedge da = 0, \quad \int_{T^3} |M_2(Du)|^2 dx = \frac{1}{16} \int_{T^3} |a \wedge Dn|^2 dx = \frac{A^2 B^2 \pi^3}{2} > 0. \quad (23)$$

Every expression depending only on da or $a \wedge da$ vanishes, whereas the vertical–horizontal shear remains strictly positive.

Second take

$$a = \sin z \, dx + \cos z \, dy + c \, dz, \quad \chi = \varepsilon \sin x. \quad (24)$$

Here $*(a \wedge da) = 1$, while $*((a + 2 \, d\chi) \wedge da) = 1 + 2\varepsilon \cos x \sin z$. The signed integrals both equal $(2\pi)^3$, but

$$\int_{T^3} |(a + 2 \, d\chi) \wedge da|^2 = (2\pi)^3(1 + \varepsilon^2) > (2\pi)^3. \quad (25)$$

Thus exactness controls the signed integral, not its squared density. $\square$

In particular the problematic AP-E13 term transforms exactly as

$$\frac{|u - q|^2}{r^2} |M_2(Du)|^2 = \frac{|u - q|^2}{16r^2} (|a \wedge Dn|^2 + |da|^2). \quad (26)$$

The counterexample disproves the *universal full cancellation*; it does not exclude a model-specific compensation theorem which also uses relaxed minimality, a Noether identity, or elimination of a defect measure.

6 Same-action finite-grid audit

The verifier re-relaxes the AP-E12 compatible-cochain action, rather than recycling stationary files. The normalized-affine map on a tetrahedron T is $u = v/|v|$, $v(\lambda) = \sum_{i=0}^3 \lambda_i u_i$. With orientation σ_T , determinant $D_T = \det[u_0, u_1, u_2, u_3]$, and physical volume V_T , the exact radial identities used in quadrature are

$$\int_T u^* \omega_3 = \sigma_T D_T \int_{\Delta} |v(\lambda)|^{-4} \, d\lambda, \quad (27)$$

$$\int_T |M_3(Du)|^2 \, dx = \frac{D_T^2}{6V_T} \int_{\Delta} |v(\lambda)|^{-8} \, d\lambda. \quad (28)$$

They reproduce the local Cauchy bound $|\int_A u^* \omega_3|^2 \leq |A| \int_A |M_3|^2$.

N	a	projected stationarity	max $\angle/a$	q99 $\angle/a$	max WZ ratio
17	0.343750	1.539×10^{-7}	3.7669	1.9147	0.8512
21	0.300000	5.003×10^{-8}	3.4636	1.8115	0.9481
25	0.270833	3.957×10^{-8}	3.3162	1.7043	0.9723

Table 1: All three backgrounds are independently relaxed, admissible, and have the three normalized-affine degree evaluations $[1, 1, 1]$. The last column is the largest local ratio $\Phi^2/(|A| \int_A |M_3|^2)$ among the reported $2a$ and $3a$ balls.

At centre-node radii $r = a, 2a, 3a$, the maximum target-angle quotients are $(2.202, 2.054, 1.915)$, $(1.964, 1.912, 1.849)$, and $(1.878, 1.846, 1.814)$ for $N = 17, 21, 25$, respectively. At $r = 2a$ the measured Φ^2/r^3 decreases $260.31 \rightarrow 150.09 \rightarrow 101.47$. These trends support a regular continuum representative, but a three-grid bounded edge quotient is not a uniform continuum Lipschitz estimate and cannot promote (8).

The production card evaluates 128 random tangent jets. Its largest residual is 8.53×10^{-14} in (16); the largest local WZ Cauchy ratio is 0.972327697. All twelve algebraic, analytic, topological, numerical, and fail-closed gate checks pass.

7 Gate decision and the next theorem

The results separate three claims which AP-E13 had left coupled:

1. Fixed-representative WZ-flux decay is **proved** and needs no variational input.
2. Linear-scale Morrey oscillation for the selected relaxed minimizer is **open**; graph bounds, degree, and WZ decay alone are insufficient.
3. A universal Hopf exact-div-curl reduction of the full positive cutoff remainder is **disproved**. Only the signed Chern–Simons part is exact.

The blocker-changing continuation is therefore not another reformulation of the WZ flux. It is a theorem about the lower-semicontinuous relaxation itself:

1. derive an inner-variation or monotonicity identity for the relaxed energy measure, allowing an explicit stress/defect measure;
2. prove that its rank-one vertical–horizontal Hopf defect vanishes, or construct a degree-preserving local comparison which removes it;
3. obtain an ε -regularity decay estimate and then upgrade mean Morrey decay to the L^∞ oscillation in (8).

Only after local recovery, classicality, and continuum isolation are proved may the same-action Riemann Hessian and determinant variation be assembled. The degree-one portal remains later still. Recent general Sobolev screening results [2] do not presently control the L^2 complete-minor endpoint required here.

Reproducibility

Run

```
python3 route_f/code/verify_ap_e14_wz_morrey_hopf_divcurl.py
```

from the repository root. The machine-readable outputs are

- route_f/output/ap_e14_wz_morrey_hopf_divcurl.json,
- route_f/output/ap_e14_wz_morrey_hopf_divcurl.md.

Production reports 12/12;wz=True;morrey=False;divcurl=False;hessian=False.

References

- [1] David Auckly and Lev Kapitanski. The pontryagin–hopf invariants for sobolev maps. *Communications in Contemporary Mathematics*, 12(1):121–181, 2010.
- [2] Pierre Bousquet, Augusto C. Ponce, and Jean Van Schaftingen. Generic topological screening and approximation of sobolev maps. *arXiv preprint*, 2025.
- [3] André Guerra, Xavier Lamy, and Konstantinos Zemas. Global minimality of the hopf map in the faddeev–skyrmie model. *Communications in Mathematical Physics*, 2026.
- [4] André Guerra, Xavier Lamy, and Konstantinos Zemas. Some lifting and approximation properties for maps in $W^{1,2}(B^3; \mathbb{S}^2)$. *arXiv preprint*, 2026.